


alvynew@gmail.com

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## TECHNICAL SKILLS

Programming in:

- C & C++
- Dart (Flutter)
- Python
- Javascript

Embedded Systems with:

- Bare-metal
- RTOS environments
- Embedded Linux

Android, iOS, web & Desktop app development using Flutter.

## INTERESTS

Programming

Internet of Things

Puzzle games

Video games

Cycling

# ALVYNE WAMWEA MWANIKI

## Embedded Firmware Engineer

Results-driven Engineer with expertise in building high-reliability distributed embedded systems, combining a meticulous attention to detail with a growth-oriented mindset and a self-motivated drive to solve complex architectural challenges.

## PROFESSIONAL EXPERIENCE

### Associate Satellite Engineer at SayariLabs

August 2022 – August 2023 (Embedded Firmware Engineer)

- ✓ Developed low-level **bare-metal drivers** for ARM cortex-M micro-controllers, using interrupt-driven and DMA-controlled approaches using C.
- ✓ Implemented inter-micro-controller **communication protocols** using UART, RS485, RS422, I2C, SPI, and USB2FS.
- ✓ Developed and tested drivers for external hardware including QSPI & SPI memory chips, IMU sensors, temperature sensors, RF transceivers, BMS balancing chips, and power monitoring chips.
- ✓ Integrated an in-house developed subsystem (SayariLabs Payload Processor) with a third-party satellite.

August 2023 – Present (On-Board Computer Subsystem Lead)

- ✓ **Architected** and spearheaded the end-to-end Flight Software development, independently developing the firmware of all satellite-bus subsystems.
- ✓ Directed the hardware-software integration and **bring-up** of custom PCBs: performing initial power-on validation, verifying peripheral and hardware functionality, and collaborating closely with hardware engineers to diagnose and resolve critical design issues in order to deliver high-reliability systems.
- ✓ Integrated custom drivers with an **RTOS** and robust error-handling with **watchdog** management to ensure maximum system up-time.
- ✓ Developed and executed **comprehensive test plans** for initial board bring up and developed firmware to ensure system reliability.
- ✓ Developed custom Python scripts and Flutter apps for **monitoring and profiling** the performance of systems in order to ensure deterministic system behaviour.
- ✓ Designed and implemented a high-reliability low-power **RF communication** stack for satellite telemetry and telecommand subsystem.

## PROJECTS

### UHF Ground Station Implementation

November 2021 - May 2022

- ✓ Undergraduate FYP (Final Year Project).
- ✓ Investigated, designed, implemented and operationalized a VHF/UHF ground station for satellite communication.
- ✓ Designed and developed a ground station control software using Visual C++, for telemetry display, telecommand issuing and image data download.
- ✓ Integrated the ground station control software with an open-source third-party software TNC (Terminal Node Controller) and to a radio, and successfully established a communication link to a nanosatellite bench model.
- ✓ Demoed the project during the first Kenya Space Expo in June 2022.

### Gearbox Kenya Internship

July 2021 - October 2021

- ✓ Spearheaded the development of the firmware of an IoT-enabled Access Control System using Atmega and ESP8266. The system's features included pass-code access, key-card access, proximity sensing, and cloud logging.

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## SOFT SKILLS

Analytical Thinking ★★★★★

Working Under Pressure ★★★★☆

Communication Skills ★★★★★

Self-discipline ★★★★★

Time management ★★★★☆

Team player ★★★★★

Team leader ★★★★☆

Implemented and tested a fully-functional prototype of the system.

Nano-Satellite Platform University of Nairobi (NaSPUoN) development
October 2020 – September 2021

- ✓ Raspberry Pi based CubeSat bench-model developed in UoN in partnership with KSA (Kenya Space Agency).
- ✓ Led the Electrical team throughout the development, from the initial satellite design, to the timely delivery of the final system back to KSA.
- ✓ Owned the development of the Flight Software and RF Communications subsystem.
- ✓ Programmed, implemented and tested the end-to-end satellite communication stack including the Ground station software (Visual C++ & .NET).

## ACHIEVEMENTS AND CERTIFICATES

University of Nairobi Certificate of Merit for Best Fourth Year Male Student in Electrical and Information Engineering in the 2020/2021 Academic Year
May 2022

IEEEXtreme 15.0 Programming Competition (Team: NameOfTeam)
October 2021 — Finishing Position: 1st in Kenya, 32nd in region R8, 103rd globally

University of Nairobi Certificate of Merit for Best Second Year Male Student in Electrical and Information Engineering in the 2018/2019 Academic Year
July 2021

IEEEXtreme 14.0 Programming Competition (Team: TitanZ)
October 2020 — Finishing Position: 1st in Kenya, 33rd in region R8, 119th globally

Hands-on Education Program for Technical Advancement (HEPTA SAT) Training
October 2019

## EDUCATION

Course: Full Stack Software Engineering | Back-end
ALX
November 2022 – March 2024 — Score: 109% (Extra is from optional tasks)

Degree: Bachelor of Science in Electrical and Electronics Engineering
University of Nairobi — GPA: 3.87/4.0
September 2017 – September 2022 — First Class Honors

KCSE High School Certificate
Strathmore School — GPA: 3.97/4.0
2013 – 2016 — Mean Grade: A- (80/84)

## REFEREES

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Eng. Bill Mbogho (Character Referee) — mboghobill16@gmail.com, (+254) 0704533821